

Xiaoqing PAN

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SUMMARY

PhD researcher in RNA biology and epitranscriptomics with background in transcriptome-wide RNA sequencing analyses, A-to-I RNA editing, alternative splicing, and small RNA biology. Experienced in integrating molecular biology with computational genomics to investigate RNA-mediated gene regulation across host and fungal systems. Proficient in RNA-seq, sRNA-seq, transcriptomic pipeline development, R, Python, Linux, and reproducible HPC workflows. Published five peer-reviewed papers and experienced in leading collaborative multidisciplinary research projects.

TECHNICAL SKILLS

- **Bioinformatics & Transcriptomics:** RNA-seq, sRNA-seq, differential expression analysis (DESeq2), Alternative splicing analysis (rMATS), A-to-I RNA editing analysis, RNA structural analysis (SHAPE-seq), Transcriptome-wide data analysis and visualization
 - **Molecular Biology & Microbiology:** RNA/DNA extraction, quantification, qPCR, CRISPR/Cas 9, Western/Northern blot, sequencing preparation, immunostaining, LC-MS, Gibson Assembly Cloning
 - **Language:** Chinese (native), English (fluent), German (B1)
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EXPERIENCES

Host Epitranscriptome and RNA Editing Analysis during fungal challenges

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differentially expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Conducted comparative genomic and transcriptomic analyses to investigate pathogen biology using reproducible computational workflows

Small RNA Biology and Transcriptome-wide RNA Regulation in *Aspergillus fumigatus*

- Developed computational workflows to characterize transcriptome-wide small RNA populations across fungal developmental stages and RNAi mutant strains.
- Performed comparative analysis of mRNA-seq and sRNA-seq datasets to identify regulatory pathways associated with fungal development and stress responses.
- Integrated sequencing analyses with experimental validation (Northern blotting and qPCR) to characterize pathogen-specific RNA regulatory mechanisms.
- Coordinated analyses within a large international collaboration involving multiple fungal genomics datasets.

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Processed extracellular RNA-seq datasets from human neutrophils following fungal challenge
- Investigate extracellular RNA dynamics and host immune RNA responses
- Contributed to collaborative transcriptomic analyses of innate immune signaling

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026

Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021

Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3*, *OsNPF2.4*, and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Riviuccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMyKG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)